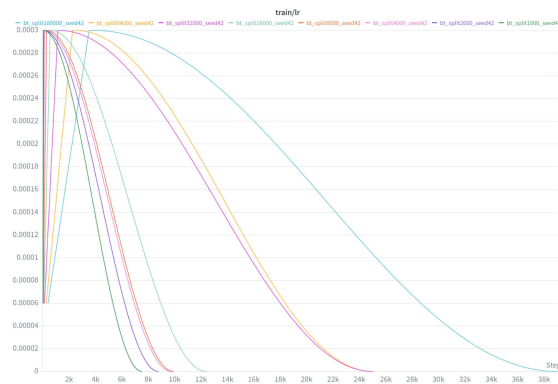
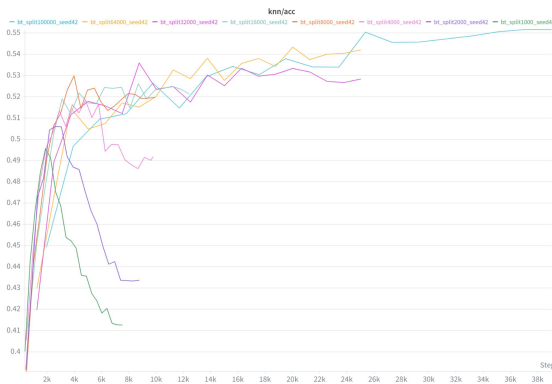
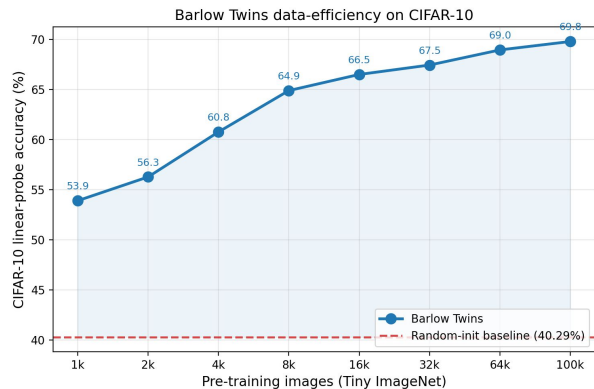


Recap

- Fixed epoch/budget overtrains for small datasets and gains marginal benefits for larger datasets. Learning rate warms up and then cosine decays.
- Epoch count is tricky to calibrate

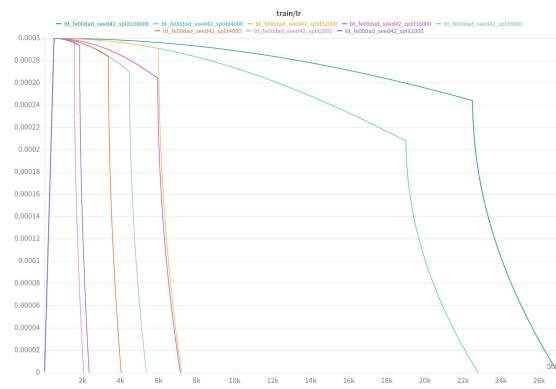
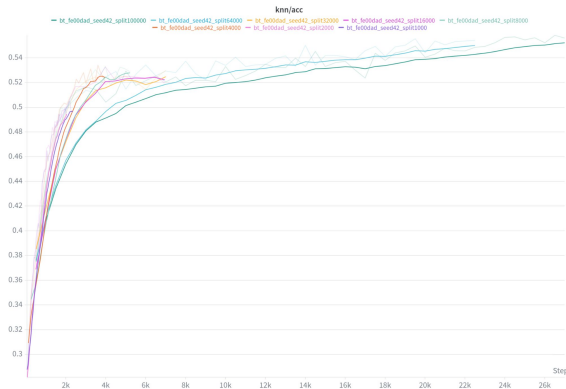
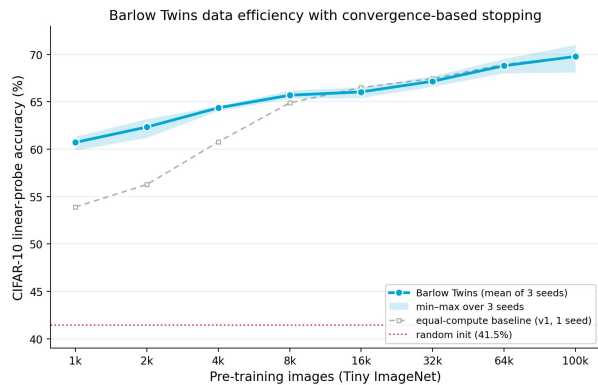


Early stopping strategy

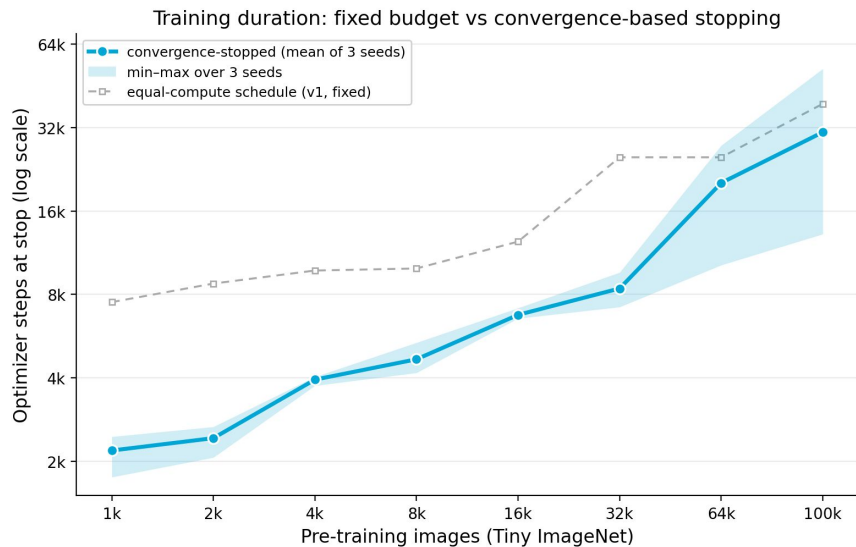
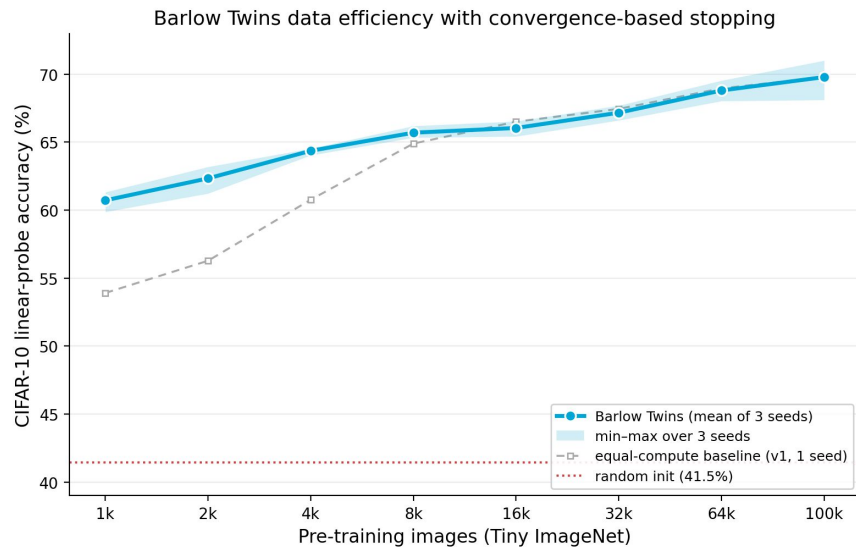
- For small datasets we want to stop as soon as validation accuracy starts to degrade – otherwise the encoder learns specific images and not patterns. For larger datasets we get diminishing returns the longer we run.
- How to know when to stop? We can use validation: when the probe accuracy hasn't increased for X consecutive probes - aggressively decay LR and stop training.
- Still tricky to calibrate our “patience” for every dataset size, but at least the stopping criteria are explainable

New runs

- All runs stop around the validation maximum
- Big improvement for small datasets, small regression for 16k (stopped too early), similar results for larger datasets



Convergence-based stopping vs Fixed epoch/budget



Next steps?

- Accuracy roughly grows with the log of the dataset size
- Large datasets can train for a long time and still improve, while small datasets overtrain quickly – can we increase the augmented image space to have 1k train and improve as 100k does?
- One concern is that harsh augmentations won't converge. What if we split training in steps and move to harder augmentations once the current ones converge? Have to run experiments to find out
- We can either add new augmentations (masking/stretching/blurring) or increase current augmentation hyperparameters (distort colors further, solarize) – will anything work? Have to experiment

First experiment: double the color distortion hyperparams (brightness/contrast, saturation, hue)

- 1k dataset, no early stopping, fixed 2000 epoch cosine decay
- Learning slows down initially but reaches a higher peak later on
- Only validation scores, no linear probing yet

